

Jun-Ting (Tim) Hsieh

juntingh@cs.cmu.edu
<https://jthsieh.github.io/>

EDUCATION

Carnegie Mellon University

Aug. 2020 - Present

Ph.D. in Computer Science
Advisor: Pravesh K. Kothari

Stanford University

Sep. 2017 - Mar. 2019

M.S. in Computer Science
GPA: 4.00/4.30
Advisors: Fei-Fei Li, Stefano Ermon

Stanford University

Sep. 2013 - Jun. 2017

B.S. in Computer Science with Distinction
GPA: 4.04/4.30
Advisor: Jure Leskovec

RESEARCH EXPERIENCE

CMU Theory Group

CMU, USA

Advisor: Pravesh Kothari

Aug. 2020 - Present

- Algorithms and lower bounds for CSPs.
- Sum-of-squares refutation of a system of random polynomial equations.
- Certifying solution geometry in random CSPs.

Stefano Ermon Group

Stanford, USA

Advisor: Stefano Ermon

Jun. 2018 - Dec. 2018

- Learn a fast neural solver for PDEs that has convergence guarantees.
- The correct solution is theoretically guaranteed to be a fixed point of the learned iterative solver.
- Our model generalizes to different geometries and boundary conditions.

Stanford Vision and Learning Lab

Stanford, USA

Advisors: Fei-Fei Li, Juan Carlos Niebles

Sep. 2017 - Mar. 2019

- Research in computer vision and deep learning.
- Video prediction: decomposition and disentanglement of videos to predict future frames.
- Action recognition and detection: graph-distillation on multi-modal privileged information.

Network Analysis

Stanford, USA

Advisor: Jure Leskovec

Mar. 2016 - Sep. 2017

- A longitudinal study involving 17 separate online news discussion communities.
- Optimization of the Breadth-First Search: directional optimization (bottom-up approach) for BFS.
- Parallel algorithms for BFS.

Institute of Atomic and Molecular Sciences

Taipei, Taiwan

Advisor: Jr-Min Lin

Jun. 2014 - Sep. 2014

- Paper published in *Science*, DOI: 10.1126/science.1261549.
- Studied the reaction of the simplest Criegee Intermediate with water.
- Data analysis to infer the amount of different chemicals from ultraviolet absorption spectrum.

PUBLICATIONS

THEORETICAL COMPUTER SCIENCE

- **Polynomial-Time Power-Sum Decomposition of Polynomials.**
Mitali Bafna, **Jun-Ting Hsieh**, Pravesh K. Kothari, Jeff Xu.
Preprint, 2022.
- **Approximating Max-Cut on Bounded Degree Graphs: Tighter Analysis of the FKL Algorithm.**
Jun-Ting Hsieh, Pravesh K. Kothari.
Preprint, 2022.
- **Certifying solution geometry in random CSPs: counts, clusters and balance.**
Jun-Ting Hsieh, Sidhanth Mohanty, Jeff Xu.
Computational Complexity Conference (CCC), 2022.
- **Algorithmic Thresholds for Refuting Random Polynomial Systems.**
Jun-Ting Hsieh, Pravesh K. Kothari.
Symposium on Discrete Algorithms (SODA), 2022.

MACHINE LEARNING & ARTIFICIAL INTELLIGENCE

- **Learning Neural PDE Solvers with Convergence Guarantees.**
Jun-Ting Hsieh, Shengjia Zhao, Stephan Eismann, Lucia Mirabella, Stefano Ermon.
International Conference on Learning Representations (ICLR), 2019.
- **Learning to Decompose and Disentangle Representations for Video Prediction.**
Jun-Ting Hsieh, Bingbin Liu, De-An Huang, Li Fei-Fei, Juan Carlos Niebles.
Conference on Neural Information Processing Systems (NeurIPS), 2018.
- **Graph Distillation for Action Detection with Privileged Information.**
Zelun Luo, **Jun-Ting Hsieh**, Lu Jiang, Juan Carlos Niebles, Li Fei-Fei.
IEEE European Conference on Computer Vision (ECCV), 2018.
- **Computer Vision-based Descriptive Analytics of Seniors' Daily Activities for Long-term Health Monitoring.**
Jun-Ting Hsieh*, Zelun Luo*, Niranjan Balachandar, Serena Yeung, Guido Pusiolo, Jay Luxenberg, Grace Li, Li-Jia Li, N. Lance Downing, Arnold Milstein, Li Fei-Fei.
Machine Learning for Healthcare (MLHC), 2018.

PHYSICAL CHEMISTRY

- **Direct kinetic measurement of the reaction of the simplest Criegee intermediate with water vapor.**
Wen Chao, **Jun-Ting Hsieh**, Chun-Hung Chang, Jim Jr-Min Lin.
Science, 2015; Volume 347, Issue 6223, page 751-754.

WORK EXPERIENCE

Lightspeed Venture Partners

Summer Fellow

Menlo Park, USA

Jun. 2017 - Sep. 2017

- *Spotlight*, an app that makes golf coaching accessible to everyone.
- Used computer vision to analyze golf swing videos.
- Built a server to process golf swing videos and give advice and suggestions to users.

Google Inc.

Software Engineer Intern

Mountain View, USA

Jun. 2016 - Sep. 2016

- Worked on Procella, YouTube's data analytics SQL engine.

- Implemented support for user-defined functions, allowing users to use custom functions in a SQL query.
- Added support for many scalar and aggregate functions in Procella.

MobiSocial Inc.

Software Engineer Intern

Mountain View, USA

Jun. 2015 - Sep. 2015

- iOS and Android development, including UI design and SQL database access.
- Integrated an external game engine to enable gaming.
- Built APIs on the server side in Java.

HONORS AND AWARDS

- **Frederick Emmons Terman Engineering Scholastic Award, 2017.**
Top 5% of the undergraduate engineering class at Stanford University.
- **Phi Beta Kappa Honor Society Member, 2017.**
Top 10% of the entire graduating class at Stanford University.
- **Stanford President's Award for Academic Excellence in the Freshman Year, 2014.**
Top 3% of the class in the freshman year.
- **Gold Medalist at International Physics Olympiad (IPhO), 2012.**
Ranked 8th overall.
- **Gold Medalist at Asian Physics Olympiad (APhO), 2012.**
Ranked 10th overall.
- **Taiwan and China High School Mechanics Competition in Beijing, China, 2013.**
Gold Medal in Theory. Gold Medal in Experiment.